Shoumik Saha

shoumiksaha.github.io smksaha@umd.edu in/shoumik-saha /ShoumikSaha

RESEARCH INTEREST

- **Machine Learning** - Security & Reliability of GenAI, LLM Alignment & Hallucination
- **Computer Security** - Adversarial Attacks & Defenses, Malware

EXPERIENCE

Applied Scientist Intern — **May 2025 – Present**
Amazon AWS — *Full Time*
Manager - Zijian Wang

Graduate Research Assistant — **Aug 2023 – Present**
Department of Computer Science — *Full Time*
University of Maryland
Supervisor - Dr. Soheil Feizi

Applied Scientist Intern — **Jun 2024 – Aug 2024**
Amazon AWS — *Full Time*
Managers - Zijian Wang, Varun Kumar

Graduate Research Assistant — **Aug 2022 – Jul 2023**
Maryland Cybersecurity Center — *Full Time*
Supervisor - Dr. Tudor Dumitras

Lecturer — **Jul 2021 – Jul 2022**
Department of Computer Science and Engineering — *Full Time*
United International University

Research Assistant — **Mar 2021 – Jul 2022**
Data Science and Engineering Research Lab — *Part Time*
Bangladesh University of Engineering and Technology (BUET)

EDUCATION

University of Maryland - College Park (UMD) — **Aug 2022 – Present**
Ph.D. in Computer Science (Ongoing) — *CGPA – 3.814/4.0*
Advisor – Dr. Soheil Feizi

University of Maryland - College Park (UMD) — **Aug 2022 – Aug 2024**
M.S. in Computer Science
Advisor – Dr. Soheil Feizi

Bangladesh University of Engineering and Technology (BUET) — **Mar 2016 – Feb 2021**
B.Sc. in Computer Science and Engineering — *Advanced GPA – 3.79/4.0*
Thesis supervisor – Dr. Atif Hasan Rahman

PUBLICATIONS

- **Almost AI, Almost Human: The Challenge of Detecting AI-Polished Writing**
  [**Shoumik Saha**, S Feizi]
  ACL 2025 (*Association for Computational Linguistics*) (Paper Link)
  Media Coverage: The New York Times, Plagiarism Today, The Science Archive, The Cheat Sheet
  We explored the growing difficulty of the detection between purely-human-written text and AI-polished text. We published the APT-Eval dataset containing 15k AI-polished texts of different degrees. We identified potential pitfalls in cases of minimally AI-polished texts.

- **Adversarial Paraphrasing: A Universal Attack for Humanizing AI-Generated Text**
  [Y Cheng, VS Sadasivan, M Saberi*, **Shoumik Saha***, S Feizi]
  Under Review (arxiv)

  We introduced Adversarial Paraphrasing, a training-free attack framework that universally humanizes any AI-generated text to evade detection more effectively. Our approach leverages an off-the-shelf instruction-following LLM to paraphrase AI-generated content under the guidance of an AI text detector, producing adversarial examples that are specifically optimized to bypass detection.

- **LLM-Check: Investigating Detection of Hallucinations in Large Language Models**
  [G Sriramanan, S Bharti, VS Sadasivan, **Shoumik Saha**, P Kattakinda, S Feizi]
  NeurIPS 2024 (*Conference on Neural Information Processing Systems*) (Paper Link)

  We introduced efficient techniques that analyze internal states, attention maps, and output probabilities to detect hallucinations from a single response, significantly improving detection performance while being less computationally expensive than previous methods.

- **Fast Adversarial Attacks on Language Models In One GPU Minute**
  [VS Sadasivan, **Shoumik Saha***, G Sriramanan*, P Kattakinda, A Chegini, S Feizi]
  ICML 2024 (*International Conference on Machine Learning*) (Paper Link)
  Media Coverage: The Register

  We proposed a novel approach in adversarial attack on LLMs, namely BEAST, that can jailbreak, cause hallucination, and membership inference attacks. Our approach can find jailbreaking prompts within one minute under a resource-constrained setting.

- **DRSM: De-Randomized Smoothing on Malware Classifier Providing Certified Robustness**
  [**Shoumik Saha**, W Wang, Y Kaya, S Feizi, T Dumitras]
  ICLR 2024 (*International Conference on Learning Representations*) (Paper Link)

  We are the first to propose certified robustness in the domain of static malware detection from executables. We demonstrated both theoretical and empirical robustness of our proposed DRSM framework. Besides, we published a new benign dataset, named PACE.

- **MAlign: Explainable Static Raw-byte Based Malware Family Classification using Sequence Alignment**
  [**Shoumik Saha**, S Afroz, A Rahman]
  Computers & Security Journal (Paper Link)

  We proposed a novel approach, namely MAlign, incorporating concepts from Bioinformatics into Malware Security. We developed a static raw-byte-based malware family classifier with better accuracy and robustness. MAlign also provides explainability by relocating the code blocks responsible for malicious attacks.

- **ML-Based Behavioral Malware Detection Is Far From a Solved Problem**
  [Y Kaya, Y Chen, Marcus Botacin, **Shoumik Saha**, F Pierazzi, L Cavallaro, D Wagner, T Dumitras]
  SaTML 2025 (*Secure & Trustworthy Machine Learning*) (Paper Link)

  We presented a quantitative study of how sandbox traces differ from real-world ones, and how it impacts machine learning models. We identified this distribution shift and proposed a solution for ML models that boosted the TPR from 14% to 20%@1%FPR.

- **Contrastive Self-Supervised Learning Based Approach for Patient Similarity: A Case Study on Atrial Fibrillation Detection from PPG Signal**
  [**Shoumik Saha***, S Shanto*, A Rahman, M Masud, ME Ali]
  Under Review (arxiv)

  We proposed Contrastive Learning based Deep Learning solution to find the patient similarity using physiological signal. We demonstrated the efficacy of our method by applying it to detect Atrial Fibrillation using PPG signals collected from smartwatches.

SCHOLARSHIPS & ACHIEVEMENTS

- Awarded *'Dean's Fellowship'* from the University of Maryland
- Awarded *'Innovation Fund'* for research from the ICT division of Bangladesh government
- Achieved *'Merit Stipends'* from BUET in five out of seven terms
- Awarded *'Dean's Award'* in Junior year for extra-ordinary result
- Achieved *'Talent-Pool Scholarship'* in High School

TEACHING

Teaching Assistant
University of Maryland — *Fall 2023*
Data 200: Knowledge in Society: Science, Data and Ethics

Lecturer
United International University — *Fall 2021 - Summer 2022*
Discrete Mathematics, Data Structure and Algorithm, Operating Systems

SERVICES

- **Reviewer**, top-tier ML conferences such as ICML 2024, ICLR 2025, NEURIPS 2025, etc.
- **External Reviewer**, top-tier Computer Security conference, including USENIX 2023
- **Lab Instructor**, United International University
- **General Secretary**, BUET Photographic Society